

Avinash Reddy Mora

Email: avinashrmora@gmail.com | **Phone:** (838) 910-9515

Location: Herndon, Virginia, USA 20171 | **Work Authorization:** H1-B

Professional Summary

- Experienced Software Engineer with 5+ years of hands-on expertise in Python backend development, RESTful API design, and cloud-native architectures.
 - Proficient in building scalable microservices using AWS (Lambda, ECS, EKS, S3), Kubernetes, and Docker.
 - Strong background in AI/ML model integration, data processing pipelines, and infrastructure automation with Terraform, Jenkins, and GitHub Actions.
 - Skilled in DevOps, CI/CD, observability (CloudWatch, Datadog, Splunk), and system performance optimization.
 - Adept at product-centric development, software architecture, and delivering end-to-end solutions in Agile environments.
 - Passionate about leveraging modern technologies to build secure, reliable, and high-performance backend systems.
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Professional Experience

SMK Soft – Herndon, VA, USA

Jan 2025 – Present

Role: IT Engineer

- Developed, built, and deployed multi-tiered software applications, utilizing DevOps tools for design, management, and troubleshooting while ensuring efficient cloud integration and application performance.
- Implemented CI/CD pipelines using Jenkins, automated infrastructure deployment with Terraform and AWS CloudFormation, and facilitated the creation of S3 buckets for template storage in the AWS environment.
- Monitored and configured application dashboards for performance tracking, performed end-to-end testing, and troubleshoot analytical and business intelligence applications to ensure seamless operation and security.

Amazon Web Services – Herndon, VA, USA

Aug 2022 – Nov 2024

Role: Software Development Engineer

- Designed, deployed, and managed Octane packages and pipelines using Amazon Apollo for continuous software delivery. Built custom version filter instances (VFIs) and integrated patching workflows to manage host updates and SLA compliance.
- Developed scalable Python packages to automate host patching and upgrade workflows, integrating PVRE APIs, host version checks, and kernel updates. Implemented multithreading, argparsers, and Slack API for notifications and execution summaries.

- Created UI workflows for rack shutdown request automation. Integrated approval flows with T.corp, MCM, DynamoDB, and Metrics, including WatchTicket-based health verification and rack recovery alarms.
- Designed red/yellow host patching workflows based on SLA and region availability using Amazon Mechanic Tool, API integrations, and AWS-hosted dashboards.
- Enhanced tooling for Garfield EC2-based storage platform and Line of Control (LoC) integrations for proactive host risk assessment during maintenance.
- Migrated host data during maintenance using automated scripts and integrated runlog monitoring with DynamoDB and Amazon QuickSight for visualization.

Cloud Infra IT Solutions – Farmers Branch, Texas, USA

May 2022 – Aug 2022

Role: Software Engineer Intern

- Performed comprehensive analysis of user needs and system requirements, applying SDLC principles to outline software specifications and guide technical design decisions using tools like Git and Agile documentation workflows.
- Designed and developed responsive user interfaces using HTML, CSS, and Bootstrap while also architecting relational database models; implemented backend logic using Java, REST APIs, and SQL to support scalable and efficient data processing.
- Conducted extensive testing and debugging of Java applications with JUnit and Postman, ensuring optimal performance, security compliance, and cross-platform functionality through both automated and manual testing practices.

Elonex Infotech Pvt. Ltd. – Hyderabad, India

May 2018 – Nov 2020

Role: Software Developer

- Migrated a legacy system to a cloud-based architecture using React.js and Node.js, improving performance and scalability while reducing downtime by 40%. Led the transition to cloud technologies, leveraging AWS for infrastructure and containerization, which significantly enhanced system reliability and resource management.
 - Refactored the frontend and integrated RESTful APIs for seamless backend communication. Dockerized and deployed the application using CI/CD pipelines on AWS, ensuring faster deployment cycles and better scalability.
 - Designed CI/CD pipelines using Jenkins and GitHub Actions with automated testing tools like JUnit and Selenium, improving development efficiency by 30% and enabling continuous integration and delivery in a robust development environment.
 - Collaborated within Agile teams to deliver full-stack applications, implementing Spring Boot APIs and PostgreSQL databases to ensure high availability and performance.
 - Developed secure and optimized database architectures using Oracle stored procedures, encryption, and RBAC, enhancing performance by 25% while ensuring data integrity and compliance.
 - Integrated real-time logging and error tracking with Log4j, improving maintainability, fault diagnosis, and reducing downtime by streamlining error management processes.
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Technical Skills & Tools

- **Programming Languages & Frameworks:** Python, Java, JavaScript, NodeJS, Go, Bash, Shell Scripting, Spring, Hibernate, React, Angular, Bootstrap, TypeScript, Perl.
 - **Architecture & Development:** Event-Driven Architecture, Application Architecture, Data Architecture, Infrastructure Architecture, Microservices, Serverless Design, REST, SOAP, JSON, Maven.
 - **Cloud & Platforms:** AWS (ECS, EKS, Lambda, S3, EC2, CloudWatch, DynamoDB, RDS, CodeBuild, CodeDeploy), Azure, Kubernetes, Docker, Cloud Foundry, Amazon Apollo, Octane.
 - **DevOps & CI/CD:** Jenkins, GitHub Actions, AWS CodePipeline, Terraform, GitOps, Helm, Maven, Brazil Build System, CI/CD, Git.
 - **Monitoring & Logging:** Dynatrace, Splunk, Datadog, Tableau, AWS CloudWatch, Log4j.
 - **Databases:** PostgreSQL, MySQL, DynamoDB, Oracle, SQL, MongoDB.
 - **Testing & QA:** Postman, JUnit, Cucumber, Selenium, Automated Testing, Manual Testing, Functional Testing, Regression Testing, Performance Testing, Unit Testing, Smoke Testing, White Box Testing, Blackbox Testing, System Integration Testing, User Acceptance Testing.
 - **Messaging & Event Streaming:** Apache Kafka, Amazon SQS, SNS.
 - **Other Tools & Technologies:** AWS QuickSight, T.corp, MCM, PVRE API, Mechanic Tool (Amazon), SDLC, Agile, Scrum, Slack API, Argparser, Multithreading.
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Education

Master of Science in Computer Science

State University of New York at New Paltz, New York, USA

Jan 2021 – May 2022

Relevant Coursework: Computer Systems, Database Design, Programming & Data Structures, Data Science, Financial Statement Analysis, Machine Learning, Artificial Intelligence, Management Operations and Information Technology, Software Engineering, Advanced Operating Systems.

Project: Sentimental Analysis

- Sentiment analysis is a natural language processing (NLP) technique used to determine whether data is positive, negative or neutral.
 - Data set has been taken from Kaggle, Number of Records in dataset: 3150 Amazon customer reviews.
 - NLP is a field in machine learning with the ability of a computer to understand, analyze, manipulate, and potentially generate human language.
 - NLP algorithms are used to provide automatic summarization of the main points in a given text or document.
 - NLP technique used to determine whether data is positive, negative or neutral.
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